Algal Research 28 (2017) 192–199

Contents lists available at ScienceDirect

Algal Research

journal homepage: www.elsevier.com/locate/algal

Effects of pH on cell growth, lipid production and CO_2 addition of microalgae *Chlorella sorokiniana*

Renhe Qiu[a], Song Gao[b], Paola A. Lopez[c], Kimberly L. Ogden[a,*]

[a] Department of Chemical and Environmental Engineering, The University of Arizona, Tucson, AZ 85721, USA
[b] Department of Agricultural and Biosystems Engineering, The University of Arizona, Tucson, AZ 85721, USA
[c] Department of Bioengineering, University of California, Berkeley, CA, USA

ARTICLE INFO

Keywords:
Microalgae
Biodiesel
Chlorella sorokiniana
pH
Lipid
Carbon dioxide

ABSTRACT

Microalgae have emerged as one of the most promising alternative energy feedstocks. Some advantages include the simple cellular structure, short production cycle, high lipid content, and fast growth. However, high production costs due to high CO_2 usage and low lipid productivity have been some of the major challenges impeding the commercial production of algal biodiesel. Here, cell growth and lipid content of *Chlorella sorokiniana* DOE1412 were first evaluated at different pH in flask cultivation. Culture pH was manipulated by CO_2 addition. The optimal pH for DOE1412 is approximately 6.0 when only accounting for cell growth and lipid production and not considering the CO_2 efficiency. A flat panel airlift photobioreactor (PBR) was used for scale-up cultivation at five different pH levels (6.5, 7, 7.5, 8 and 8.5). Data of pH values and CO_2 addition was collected by a data logger. Biomass productivity increased with decreasing pH. By taking into account not only the cell growth and lipid production but also CO_2 addition, the lowest value of CO_2 addition was achieved at pH 8 (2.01 g CO_2/g biomass). The fatty acid profiles and biodiesel properties, such as iodine value (IV), saponification value (SV), cetane number (CN), degree of unsaturation (DU), long-chain saturated factor (LCSF), and cold filter plugging point (CFPP), were determined as a function of pH. CN of biodiesel produced at pH 6.5, 7 and 7.5 satisfied the US standard ASTM D6751; among them, the pH 6.5 products met the European standard EN 14214. Finally, protein content in microalgal biomass increased with increasing pH, while C/N ratio in cells decreased.

1. Introduction

Microalgae can fix CO_2 efficiently from industrial exhaust gasses during photosynthesis, providing a very promising alternative for the mitigation of CO_2, the most prominent greenhouse gas [1–6]. Microalgae have the ability to utilize nutrients (especially nitrogen and phosphorus) from polluted municipal, industrial and agricultural wastewater, which allows for use of non-potable water for growth and provides a promising option for the bioremediation of wastewater [7–10]. However, the production cost of microalgal biodiesel is currently not competitive with petroleum-based products. Many significant technical challenges need to be tackled before this pathway is ready for commercial-scale implementation [11]. Some example research and development areas include: studying the molecular biology of microalgae; selecting and bioengineering microalgae strains optimized to regional climate conditions; developing large-scale microalgae cultivation systems; optimizing cultivation conditions (e.g., nutrient, temperature, light, pH, salinity, and mixing); developing efficient techniques for lipid extraction, biomass harvesting, and downstream processing; reducing and controlling operating costs and energy consumption; and utilizing residues to make valuable co-products [12,13]. This work specifically focuses on optimizing the pH and the amount of CO_2 required for efficient algal growth.

Chlorella sorokiniana DOE1412, a member of the Trebouxiophyceae class, also known as UTEX B 3016, was the microalgae strain used in this study. It has been reported as an important microalga with potential for biodiesel production, accumulating high amounts of lipids and proteins [14,15]. Moreover, it can grow robustly at temperatures of up to 40 °C and in impaired water [5,16].

Microalgae cultivation can be optimized by regulating a variety of environmental parameters, such as light, temperature, nutrients, salinity, temperature, and pH [17,18]. These parameters not only affect the accumulation of biomass profoundly but also influence the biochemical composition of microalgal biomass. The pH is one of the most critical environmental conditions in microalgal cultivation since it determines the solubility and availability of CO_2 and nutrients, and has a significant influence on microalgal metabolism [19]. Each microalgal species has an optimal pH range for biomass and lipid production,

* Corresponding author.
E-mail address: ogden@email.arizona.edu (K.L. Ogden).

http://dx.doi.org/10.1016/j.algal.2017.11.004
Received 29 May 2017; Received in revised form 1 November 2017; Accepted 3 November 2017
Available online 08 November 2017
2211-9264/ © 2017 Published by Elsevier B.V.

which is narrow and strain specific [20–23]. Specifically, pH is critical in determining the relative concentrations of the carbonaceous species in water [24]. The pH of microalgal cultures rises gradually during the day due to the uptake of inorganic carbon by microalgae. Higher pH limits the availability of CO_2, thus, inhibiting cell growth [19,24]. On the other hand, algal cultivation at high pH can suppress undesired biological contaminants [25].

Methods for controlling pH include CO_2 injection, buffer addition, and acid/base adjustment. The former two are more commonly used in algae cultivation [20,25,26]. However, the presence of buffers can be a potential pitfall for research because, to date, effects of buffers on microalgal metabolism are not entirely clear. Also, it is not cost-effective or realistic at large scale. The present study focuses on regulating pH by controlling and measuring the amount of CO_2 injected during cultivation. CO_2 addition is a significant economic consideration for biodiesel production. However, carbon balances are not typically performed on algal bioreactors and many economic studies do not provide detailed analyses of the costs associated with producing and transporting CO_2 [27]. A common simplifying assumption is based on co-locating an algal production system with a facility that produces waste CO_2, without considering the potential negative effects associated with toxic substances in flue gas on algal growth, the availability of land near the flue gas, or the possibility for scalability. More recently, the need to understand the costing for CO_2 was highlighted by some studies [28,29]. CO_2 and land were identified as two barriers for the large-scale potential of microalgae [28].

In this study, both Erlenmeyer flasks and flat panel photobioreactor (PBR) cultivation systems were used. Moreover, real-time data of temperature, dissolved oxygen, pH and CO_2 flow rate were monitored during PBR cultivation. An optimal culturing pH range was determined by taking into account not only the cell growth and lipid production but also CO_2 addition. The implications of this research demonstrate the tradeoffs between growth and CO_2 addition that can be implemented in the field to enhance the economical and sustainable viability of algal cultivation.

2. Materials and methods

2.1. Strains and growth medium

Chlorella sorokiniana DOE1412 [30] was obtained from the National Alliance for Advanced Biofuels and Bioproducts (NAABB) cultivation team. Currently it is available from UTEX. The seed culture (500 mL in a 1 L Erlenmeyer flask) was continuously shaken on a platform at 120 rpm under a 12:12 h light/dark period at ambient temperature. Culture medium was replaced every 14 days. Algae cells were re-suspended in fresh medium after old medium was removed by centrifugation.

The standard growth medium used for freshwater *Chlorella* sp. is BG-11 [31]. For this work, an optimized medium PE-001A was used. *Chlorella sorokiniana* DOE1412 grew as well on the PE-001A medium as on the BG-11 medium; however, the PE-001A medium recipe is ten times less expensive [32]. The PE-001A medium consists of the following: 100 mg/L $(NH_2)_2CO$, 12 mg/L $MgSO_4{\cdot}7H_2O$, 25 mg/L $NH_4H_2PO_4$, 75 mg/L Potash, 3.15 mg/L $FeCl_3$, 20 mg/L Na_2CO_3, 4.36 mg/L EDTA and 1 mL/L trace metal solution. The trace elements in the solution consist of the following: 2.86 g/L H_3BO_3, 1.81 g/L $MnCl_2{\cdot}4H_2O$, 0.22 g/L $ZnSO_4{\cdot}7H_2O$, 0.39 g/L $Na_2MoO_4{\cdot}2H_2O$, 79 mg/L $CuSO_4{\cdot}5H_2O$ and 49.4 mg/L $Co(NO_3)_2{\cdot}6H_2O$. Based on pH titration of PE-001A medium, a pKa = 6.2 is observed between pH 6 and pH 9.

2.2. Flask cultivation

Chlorella sorokiniana DOE1412 was cultured in a 1 L Erlenmeyer flask using sterile PE-001A medium at room temperature (25 °C). The culture volume was 500 mL. The artificial light source consisted of four 61 cm long 54 watt fluorescent light tubes (FLT5464, Hydrofarm, Inc., Petaluma, CA) illuminating the top of the flasks, with a light intensity of 200 μmol photons $m^{-2}s^{-1}$ on the flask surface, measured by a quantum meter (MQ-200, Apogee Instruments, Logan, UT). The light cycle was 12 h-on/12 h-off. The initial biomass concentration was approximately 0.06 g/L. Cultures were grown at four different pH levels (6, 7, 8 and 9) using a pH controller to adjust injection of CO_2. Air at a flow rate of 1 L/min (LPM) was added to improve culture mixing. For the control group, a CO_2/air mixture (5/95, v/v) was continually injected into each flask at 1 LPM. All the gasses were saturated with water before entering each flask to decrease water evaporation.

2.3. PBR cultivation

Airlift flat-panel PBRs were used for scale-up cultivation. They were built using 6.35 mm thick transparent acrylic panels with dimensions of 122 (H) × 91 (L) × 10 cm (W). The working volume was 90 L. The air was continually injected into the PBR via three air bubbling stones (Discard-A-Stones 12,526, Lee's Aquarium & Pet Products, San Marcos, CA) placed at the bottom of the PBR to achieve proper mixing of the microalgae culture. An air compressor (3Z355H, W. W. Grainger, Inc., Lake Forest, IL) was used to maintain the proper injection pressure. Air flow was kept constant at 6 LPM to ensure the culture was well mixed. CO_2 was injected discontinuously (0.3 LPM when it was on) via a micro bubble diffuser (1PMBD075, Pentair Aquatic Eco-Systems, Inc., Apopka, FL), controlled by a data logger (CR1000, Campbell Scientific, Inc., UT, Logan, UT) to maintain the desired culture pH. The pH (HI1006-2005, Hanna Instruments, Inc., Woonsocket, RI), dissolved oxygen (DO6400/T, Sensorex, Garden Grove, CA) and temperature (109SS-L, Campbell Scientific Inc., Logan, UT) probes were placed in the PBR and scanned every second. The data logger was also connected to a digital mass flow meter (TopTrak 820, Sierra Instruments, Monterey, CA), to monitor CO_2 addition. The 10-minute averaged data were stored. The PBR was illuminated by eight 61 cm long 54-watt fluorescent light tubes (FLT5464, Hydrofarm, Inc., Petaluma, CA) providing 250 μmol photons $m^{-2}s^{-1}$ on the surface.

Chlorella sorokiniana DOE1412 was grown in 4 L Erlenmeyer flasks provided with a 8 LPM flow rate of a CO_2/air mixture (5/95, v/v) each, at room temperature (25 °C), for seven days before being transferred to the PBRs. The initial biomass concentration was approximately 0.3 g/L. Cultures were grown at five different pH levels (6.5, 7, 7.5, 8 and 8.5), in duplicates, in the PBRs.

2.4. Biomass concentration

The biomass concentration, in dry weight, was determined by measuring the optical density of samples at wavelength 750 nm (OD_{750}) using an ultraviolet-visible spectrophotometer (Genesys 10S UV–Vis, Thermo Fisher Scientific, Inc., Waltham, MA). Daily samples were taken from the reactors during the daylight hours. They were diluted by appropriate ratios to ensure that the measured OD_{750} values were in the linear range of 0.1–0.7. The relationship between OD_{750} and biomass concentration was determined experimentally. Biomass was centrifuged, washed, and dried. A linear regression relationship was obtained:

$$\text{Biomass concentration} = \text{conversion factor} \times OD_{750} \times \text{dilution factor} \quad (1)$$

The conversion factor was validated periodically for all the treatments.

2.5. Lipid content

Algae samples were harvested by a centrifuge (J2–21, Beckman Coulter, Inc., Brea, CA) at relative centrifugal force (RCF) of 4400 × *g* for 10 min. The pellets were dried at 37 °C in a forced air oven (414005-

114, VWR International LLC, Radnor, PA) to determine the dry biomass weight, then ground to a fine powder using a mortar and pestle.

The lipids were extracted using microwave-assisted extraction [33]. The powdered algae were transferred to MARSXpress vessels (0.3–0.5 g per vessel) of the CEM Microwave Accelerated Reaction System (MARS, CEM Corp., Matthews, NC). Ten mL of pre-mixed chloroform-methanol (2:1 by volume) were added to each vessel and held at room temperature for at least 2 h. The suspensions were then heated via microwave to 70 °C and held at 70 °C for 60 min [34]. Vessel contents were cooled to room temperature. The samples were then filtered with glass fiber and dried with a nitrogen evaporator (N-Evap 112, Organomation Association Inc., Berlin, MA). The lipid content was determined by weight (lipid/biomass, x%).

2.6. Fatty acid profiles

A transesterification reaction is necessary to convert the lipid obtained from microwave-assisted extraction into fatty acid methyl esters (FAMEs). Transesterification was done by reacting the extracted lipid with 3 M methanolic HCl [35]. The lipids and solvent were heated for 10 min at 57 °C to convert them to FAMEs [36].

The FAMEs were solubilized in dichloromethane (dilution factor of 5) and analyzed with an Agilent 7890A gas chromatograph (GC) coupled to an Agilent 5975C inert XL mass selective detector (MSD) with Triple-Axis Detector (Agilent Technologies, Santa Clara, CA). A 1 μL sample was injected in pulsed splitless mode by an Agilent 7693 autosampler (Agilent Technologies, Santa Clara, CA). Separation of compounds was accomplished on a fused silica capillary column (Omegawax 250, Supelco, Bellefonte, L × I.D. 30 m × 0.25 mm, d_f 0.25 μm). The oven temperature program was 2 min at 90 °C, increasing at 3 °C min^{-1} to 240 °C, where the temperature was maintained for 15 min. Helium was used as carrier gas at a constant flow rate of 1.0 mL/min. The solvent delay was 5 min. The mass spectrometry (MS) source temperature was set to 230 °C; the quadrupole temperature was set to 150 °C, and the transfer line temperature was 180 °C. Spectra were acquired in positive (70 eV) full scan mode from 50 to 600 m/z at 1.4 spectra/s scan speed.

2.7. Biodiesel properties

The biodiesel properties such as iodine value (IV), saponification value (SV), cetane number (CN), degree of unsaturation (DU), long-chain saturated factor (LCSF), and cold filter plugging point (CFPP) were determined by empirical equations. CN was calculated using Eq. 2 [37]:

$$CN = 46.3 + \frac{5458}{SV} - 0.225 \times IV \quad (2)$$

where SV in mg KOH/g and IV in g I_2/100 g were calculated from fatty acid profiles [38]:

$$SV = \sum_i \frac{560 \times N_i}{M_i} \quad (3)$$

$$IV = \sum_i \frac{254 \times D_i \times N_i}{M_i} \quad (4)$$

N_i is the percentage of each component, M_i is the molecular mass, and D_i is the number of double bonds.

DU was determined by an empirical equation, taking into account the weight percentage of monounsaturated fatty acids (MUFA) and polyunsaturated fatty acids (PUFA) [39]:

$$DU = MUFA + (2 \times PUFA) \quad (5)$$

LCSF and CFPP were obtained based on Eqs. 6 and 7 [39]:

$$LCSF = (0.1 \times C16) + (0.5 \times C18) + C20 + (1.5 \times C22) + (2 \times C24 \quad (6)$$

$$CFPP = (3.1417 \times LCSF) - 16.477 \quad (7)$$

where C16, C18, C20, C22 and C24 are the weight percentage of each saturated fatty acid.

2.8. Total carbon (TC) and total nitrogen (TN) content

TC and TN content in microalgae biomass was quantified by the Arizona Laboratory for Emerging Contaminants (ALEC, Tucson, AZ), using an elemental combustion analyzer ECS-4010 (Costech Analytical Technologies, Inc., Valencia, CA). All the biomass samples were dried and ground to a fine powder prior to analysis. The system used helium carrier gas to bring the combustion gasses to GC separation and a thermal conductivity detector (TCD). Sulfanilamide, NIST 1632b coal, and LECO samples including soil, corn gluten, and cysteine were used as calibration standards. Triplicate measurements were made on each sample.

2.9. Statistical analysis

One-way analysis of variance (ANOVA) was used for statistical analysis. The analyses were conducted using JMP software (JMP Pro 12.2.0, SAS Institute Inc., Cary, NC). Results were reported at 95% confidence intervals.

3. Results and discussion

3.1. Effects of pH on cell growth and lipid content in 500 mL flasks

Fig. 1 shows the growth curves of DOE1412 in flask cultivation at different pH. Data are the average of duplicates. The pH of the control group (continuous addition of 5% CO_2 in air) was 5.8 ± 0.1 during the entire growth period. After a one day lag phase, the growth curves were observed to be close to linear for the first four days. The growth and productivity data are shown in Table 1. The linear growth rates for pH 5.8, 6, and 7 were statistically the same (one-way ANOVA: $p > 0.05$), with the highest average rate (0.353 g/L-day) observed at pH 7. The growth rates decreased with increasing pH from 7 to 9. Also, the R^2 value for the linear regression was much closer to 1 at higher pH, due to CO_2 limitation. The effect of pH was investigated using CO_2 addition by Moheimani [20] for an unidentified species of *Chlorella*. The highest specific growth rate observed for microalgae grown in a 5 L open glass aquarium PBR, was at pH 7. Since their reactors were operated in a semi-continuous mode by harvesting three times a week, the data obtained was for short growth periods of 3–4 days, which is similar

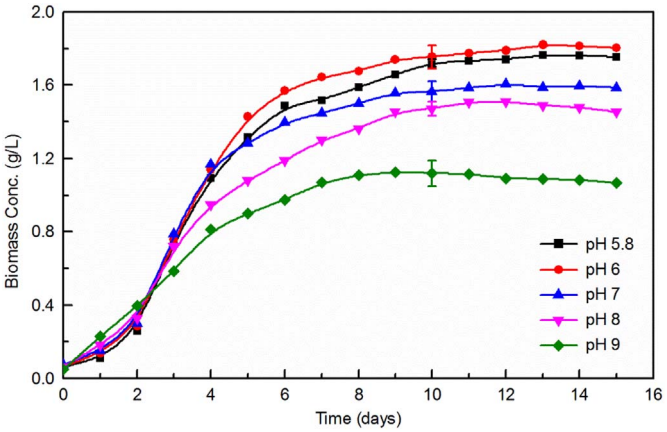


Fig. 1. Growth curves for *Chlorella sorokiniana* DOE1412 grown at different pH in 500 mL flasks. Data are the average of duplicates. ± S.D. lines at t = 10 day are shown.

Table 1
Growth and productivity summary at different pH in flasks.

pH	Linear growth rate (g/L-day)	R^2	Biomass productivity (g/L-day)
5.8	0.345 ± 0.004	0.96	0.138 ± 0.002
6	0.348 ± 0.005	0.96	0.140 ± 0.004
7	0.353 ± 0.004	0.96	0.119 ± 0.005
8	0.269 ± 0.010	0.97	0.109 ± 0.003
9	0.193 ± 0.007	0.99	0.071 ± 0.004

to our linear growth period.

The biomass productivity were calculated for a two-week culture period (Table 1), which is after stationary phase was reached. The maximum biomass productivity observed was at pH 6 (0.140 g/L-day), which was statistically the same as the productivity at pH 5.8 (one-way ANOVA: $p > 0.05$). Also, we found that biomass productivity decreased with increasing pH from 6 to 9. The productivity was significantly lower at pH 9 (0.071 g/L-day), almost half of the value for pH 6. Thus, our observed optimal pH for DOE1412 biomass productivity in batch culture was approximately pH 6.

The pKa of the PE-001A medium is 6.2 based on titration experiments. When microalgae DOE1412 were suspended in the medium, the pKa range of bicarbonate buffering system was stable. The dissolved inorganic carbon is mainly found in three forms in the system, including aqueous carbon dioxide (CO_2(aq)), bicarbonate (HCO_3^-), and carbonate (CO_3^{2-}), where CO_2(aq) and HCO_3^- can be utilized by most of the microalgae for photosynthesis. However, some algae species only use HCO_3^- to complement CO_2, while others can efficiently utilize HCO_3^- alone [40]. The distribution of inorganic carbon species is pH-dependent. At pH 9, the predominant form is HCO_3^-, while both CO_2(aq) and HCO_3^- are equally present at pH 6. As the algae grow, the total dissolved inorganic carbon decreases, and pH slowly increases. After additional CO_2 is added to the system, the pH immediately drops to the desired value, and the equilibrium between the carbon species is re-established accordingly. Based on the results of this study, DOE1412 appeared to have the ability to use HCO_3^- as a sole carbon source, however the algal growth rate was significantly lower than when both CO_2(aq) and HCO_3^- are present in the media. Lam and Lee [41] found that *Chlorella vulgaris* preferred to use CO_2(aq) at acidic conditions instead of HCO_3^- at alkaline conditions, which agrees with the findings of this study.

Fig. 2 shows the lipid content of DOE1412 grown in flasks at different pH. Lipid content appeared to be unaffected by pH (one-way ANOVA: $p > 0.05$) with an average value of 29.5% by mass. Moheimani [20] found similar results in that the lipid content of *Chlorella* sp. remained constant at 23 ± 2%. Since the species of *Chlorella* is different in the two studies, it is not surprising that the actual percentage lipid is different.

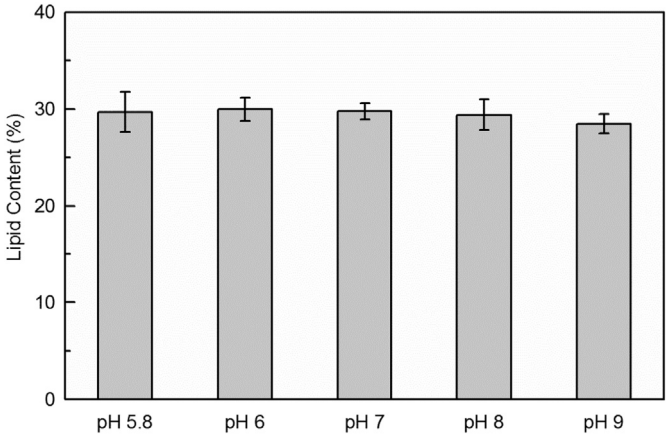


Fig. 2. Lipid content for *Chlorella sorokiniana* DOE1412 grown at different pH in 500 mL flasks. Data are shown as means ± S.D., $n = 4$.

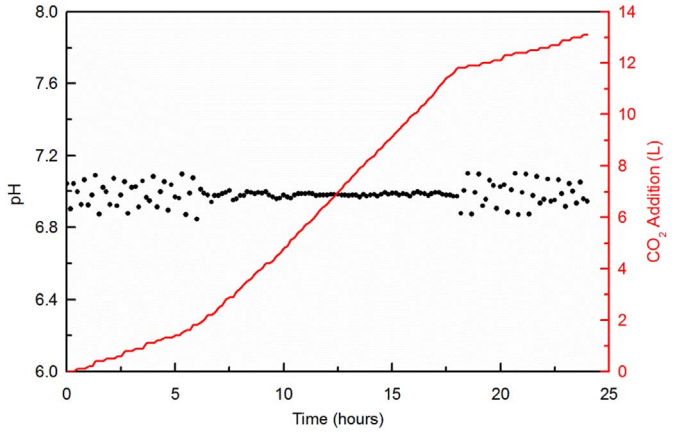


Fig. 3. pH and CO_2 addition in a 90 L PBR when pH was maintained at 7 within 24 h for *Chlorella sorokiniana* DOE1412 culture (Light: 6–18 h; Dark: 0–6 & 18–24 h).

3.2. Effects of pH on cell growth, CO_2 addition and lipid content in 90 L airlift PBRs

After determining the optimal pH range in easy to control bench scale flasks, the cultivation system was scaled up to 90 L PBRs. At the larger scale, it is easier to monitor multiple parameters simultaneously including pH, temperature, DO, and CO_2 addition. Our focus was CO_2 addition as a function of pH. The scaled-up PBR reactors were designed specifically to monitor CO_2 addition, using a data logger to control pH. Based on bench scale results, the pH range investigated was from 6.5 to 8.5. Using a pH below 6.5 is not cost-effective due to the extremely high amount of CO_2 addition (44.24 L/day), > 2.5 times the amount required at pH 7. On the other extreme, pH 9, biomass productivity was significantly lower than what we observed at pH 8.

Fig. 3 shows the culture pH and cumulative CO_2 addition during a 24-hour period, from 1:00 am to 0:59 am when the desired culture pH was 7. No initial pH adjustment of medium was performed. It took < 1 min to reach the desired pH. During the daytime (7 am to 7 pm, 6–18 h in Fig. 3), pH remained at 6.98 ± 0.01. During the nighttime, it was 6.99 ± 0.07. Both CO_2 injection and respiration during the dark growth phase lowered the pH in the PBR, which caused larger pH fluctuations. With no photosynthesis during the dark period, CO_2 addition was much less than during the daytime as expected. The small amount of addition might be due to CO_2 loss to the atmosphere.

Fig. 4 shows the growth curves of DOE1412 and CO_2 addition during PBR cultivation at different pH values. Data are the average of duplicates. DOE1412 exhibited a longer linear growth phase with lower growth rates in these larger reactors, which were light limited. The linear growth rates calculated using the first ten days of data were 0.091 g/L-day for pH 6.5, 0.074 g/L-day for pH 7, 0.068 g/L-day for pH 7.5, 0.061 g/L-day for pH 8, and 0.023 g/L-day for pH 8.5. Biomass growth rates decreased with increasing pH from 6.5 to 8.5, with a significant drop in rate at a pH of 8.5 (about a quarter of the rate at pH 6.5).

The biomass productivity and CO_2 addition as a function of pH are shown in Table 2. Biomass productivity was determined by the average biomass accumulation before the culture reached the stationary phase. It generally increased with decreasing pH, where the productivity at pH 6.5 > pH 7 > pH 7.5 > pH 8 > pH 8.5. CO_2 addition (g CO_2/g biomass) was calculated using the density of CO_2 at standard temperature and pressure (1.96 g/L). The cost of CO_2 was estimated using the current cost of existing pipeline CO_2 supplies for low pressure delivery ($15/tonne) [42]. At low pH, more CO_2 was added to the PBR, providing more carbon to the DOE1412 culture. Although excess CO_2

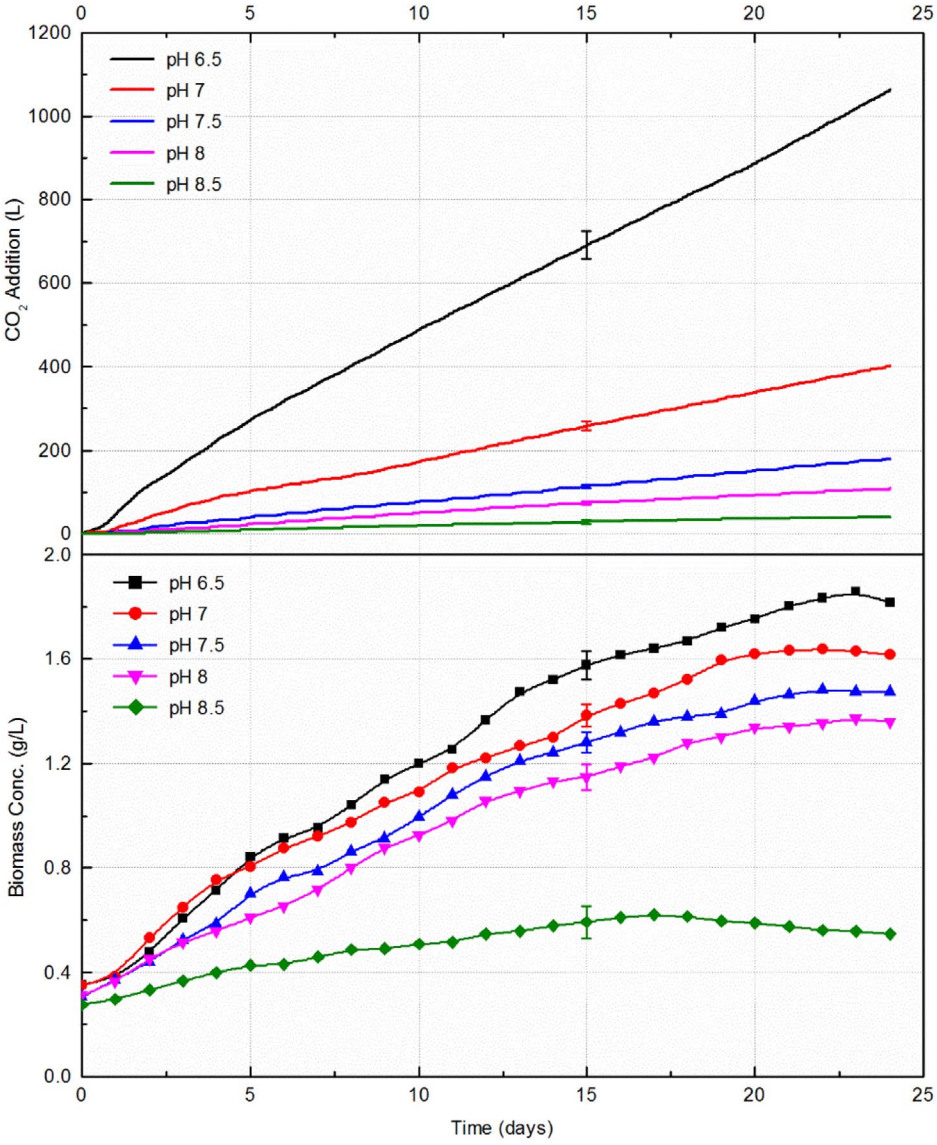


Fig. 4. Growth curves for *Chlorella sorokiniana* DOE1412 and CO_2 addition for cultures at different pH in a 90 L PBR. Data are the average of duplicates. ± S.D. lines at t = 15 day are shown.

Table 2
Biomass productivity, CO_2 addition, and CO_2 cost estimate for cultures at different pH in a 90 L PBR.

pH	Biomass productivity (g/day)	CO_2 addition			CO_2 cost estimation ($/tonne biomass)
		(L/day)	(L CO_2/g biomass)	(g CO_2/g biomass)	
6.5	6.51	44.24	6.79	13.32	200
7	5.85	17.15	2.93	5.75	86
7.5	5.07	7.71	1.52	2.98	45
8	4.62	4.73	1.02	2.01	30
8.5	1.84	1.98	1.08	2.12	32

led to higher biomass productivity, CO_2 was utilized less efficiently. The CO_2 addition (g CO_2/g biomass) decreased with increasing pH from 6.5 to 8, while the value at pH 8.5 was slightly higher the value at pH 8. The CO_2 addition was highest at 13.32 g CO_2/g biomass at pH 6.5, costing $200/tonne biomass, which might not be economically viable. The lowest value was 2.01 g CO_2/g dry biomass at pH 8, costing $30/tonne biomass, which was the most cost-effective culture pH for this algal species, in term of CO_2 addition.

Similar to the flask results, the lipid content was not a function of pH in the PBRs; no significant differences were found (See Fig. 5. one-way ANOVA: $p > 0.05$). Overall, the lipid content was slightly lower at the larger scale (25.7% by mass), which is often observed. These values were similar to the observations of Moheimani [20].

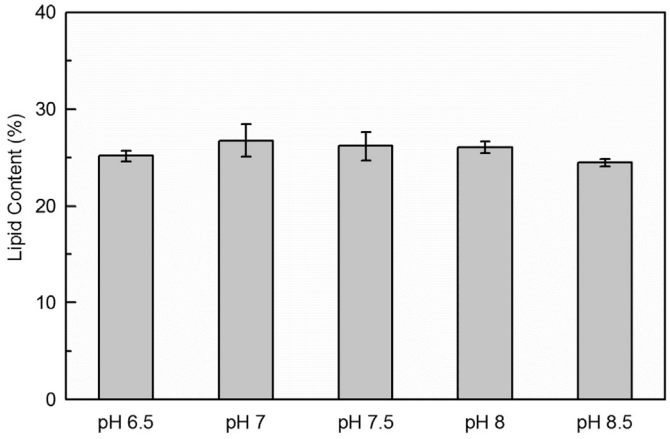


Fig. 5. Lipid content for *Chlorella sorokiniana* DOE1412 grown at different pH in a 90 L PBR. Data are shown as means ± S.D., n = 4.

Table 3
Fatty acid profiles at different pH in a 90 L PBR (% of total FAMEs). Data are shown as means ± S.D., $n = 4$.

Fatty acid	pH 6.5	pH 7	pH 7.5	pH 8	pH 8.5
Palmitic (C16:0)	39.52 ± 0.36	37.63 ± 0.25	35.92 ± 0.58	34.27 ± 0.28	29.24 ± 0.70
Palmitoleic (C16:1)	3.46 ± 0.26	2.55 ± 0.10	2.87 ± 0.44	3.72 ± 0.08	2.12 ± 0.05
Hexadecadienoic (C16:2)	6.82 ± 0.75	7.13 ± 0.28	7.69 ± 0.50	7.62 ± 0.68	10.93 ± 0.81
Hexadecatrienoic (C16:3)	2.97 ± 0.14	4.90 ± 0.27	5.66 ± 0.23	7.19 ± 0.39	8.95 ± 0.42
Stearic (C18:0)	1.20 ± 0.17	1.46 ± 0.08	1.59 ± 0.26	1.49 ± 0.16	1.14 ± 0.07
Oleic (C18:1)	13.06 ± 0.20	12.72 ± 0.54	11.93 ± 0.42	9.16 ± 0.23	8.11 ± 0.06
Linoleic (C18:2)	24.98 ± 0.50	23.39 ± 0.31	21.70 ± 0.75	22.41 ± 0.64	21.85 ± 0.25
Linolenic (C18:3)	5.16 ± 0.17	7.55 ± 0.03	9.62 ± 0.08	11.28 ± 0.39	15.25 ± 0.33
Not identified (NI)	2.84 ± 0.19	2.69 ± 0.27	3.04 ± 0.44	2.88 ± 0.37	2.44 ± 0.97
SFA	40.72 ± 0.19	39.09 ± 0.17	37.50 ± 0.33	35.75 ± 0.14	30.37 ± 0.76
MUFA	16.51 ± 0.07	15.27 ± 0.64	14.80 ± 0.05	12.88 ± 0.16	10.22 ± 0.10
PUFA	39.93 ± 0.30	42.97 ± 0.03	44.67 ± 0.13	48.50 ± 0.10	56.97 ± 1.81

3.3. Effects of pH on fatty acid profiles and biodiesel properties in a 90 L airlift PBR

It is well known that fatty acid profile determines the quality of produced biodiesel, especially when the microalgae strain is able to produce PUFAs. As shown in Table 3, DOE1412 mostly accumulated 16 and 18 carbon fatty acids, which are suitable for biodiesel production. The primary components were C16:0 and C18:2. Neofotis et al. investigated the changes in the fatty acid profile of DOE1412 during a growth period of 16 days and found similar results: C16:0 and C18:2 became predominant species after a week of growth [15]. The fatty acid profiles appeared to be influenced by culture pH. The fractions of SFA and MUFA decreased with increasing cultivation pH from 6.5 to 8.5, while the fraction of PUFA increased. Generally, unsaturated fatty acids, consisting of MUFAs and PUFAs, have lower melting points than SFAs. They are excellent for cold weather biodiesel production, but require an oxidative stabilizer to be used safely. It has been reported that MUFAs are most favorable for biodiesel production, because of the melting points that allow for flow in cold climates, while limiting the amount of oxidizer addition [43]. In this study, the highest fraction of MUFA (16.51%), consisting of C16:1 and C18:1, was observed when *Chlorella* was cultivated at pH 6.5.

Based on the fatty acid profiles, this study also estimated several important properties of biodiesel, shown in Table 4. The CN is a prime dimensionless indicator of the ignition quality of diesel, determined by the ignition delay time in a diesel engine. Higher CN leads to better ignition quality and better engine performance [44]. For the two common biodiesel standards, ASTM D6751 (United States) and EN 14214 (Europe), the minimum CN values are 47 and 51, respectively. In this work, the estimated CN values vary from 41 to 51, higher CN values are achieved when the algae are cultivated at lower pH. Fatty acids produced at cultivation pH values of 6.5, 7 and 7.5 satisfied the US standard, among them, pH 6.5 met the European standard. It was reported that increasing DU led to decreasing CN [45], which is also observed in this study. The IV is a crude measure of fatty acids unsaturation, which relates to the oxidative stability of diesel [43]. EN 14214 defines a maximum IV specification of 120 mg I_2/100 g, while ASTM D6751 does not include this specification. In this study, fatty acids produced at pH 6.5, 7 and 7.5 met the European standard, while pH 8 products were just 0.6% higher than the threshold. Finally, the CFPP is an indicator of diesel flow performance at low temperature. Better cold-weather performance is observed at lower CFPP values. Both ASTM D6751 and EN 14214 do not mention a CFPP limit; however, each country can specify certain temperature limits based on geographic location and time of year [46]. With CFPP ranges from − 5.5 °C to − 2.2 °C, the biodiesel that would be produced regardless of culture pH are predicted to have decent cold-weather performance.

Overall, biodiesel that would be produced from *Chlorella* cultivated at lower pH generally could meet the diesel standard. Furthermore, since biodiesel is often blended in the United States; a 20% blend of biodiesel and petroleum diesel should be compatible with all diesel engines [47]. Biodiesel produced from DOE1412 cultivated at pH 6.5–8.5 can satisfy both US and European standards by blending it with a relatively high CN diesel.

Table 4
Comparison of biodiesel properties as a function of cultivation pH to ASTM and EN biodiesel standards.

Property	pH 6.5	pH 7	pH 7.5	pH 8	pH 8.5	ASTM D6751	EN 14214
CN	51	49	48	46	41	≥ 47	≥ 51
SV (mg KOH/g)	204	205	204	205	205	–	–
IV (g I_2/100 g)	97	106	112	121	140	–	≤ 120
DU (wt%)	96	101	104	110	124	–	–
LCSF (wt%)	4.6	4.5	4.4	4.2	3.5	–	–
CFPP (°C)	− 2.2	− 2.4	− 2.7	− 3.4	− 5.5	–	–

3.4. Effects of pH on protein content and carbon to nitrogen (C/N) ratio in a 90 L PBR

For microalgal biomass, the protein content can be as high as 50% by weight, with a low C/N ratio of about 10 [48]. Lourenco et al. provided a practical method for determining protein content suitable for microalgae, using a nitrogen-to-protein convention factor of 4.78 [49]. Templeton and Laurens [50] reported a new conversion factor (4.08) determined from new data along with literature. In this work, the factor of 4.78 was used to calculate protein content, because it was more commonly used in other studies [51]. TC, TN, and protein content as well as C/N ratio of the DOE1412 biomass produced at different pH in PBRs are shown in Table 5. Data are the average of triplicates. TC content appeared to be unaffected by pH (one-way ANOVA: $p > 0.05$). At pH 8, 2.01 g CO_2 were added to produce each gram of biomass (Table 2).The resulting biomass consisted of 48.12% C by mass. According to the carbon mass balance, 87.8% of CO_2 added was efficiently fixed by microalgae at pH 8. Protein content, which was converted from TN content, increased with increasing pH. The values ranged from 15.5% to 30.7%, while C/N ratio in the cells decreased. Ogbonda et al. [52] also found a significant difference in protein production of *Spirulina* sp. at various pH, with the most protein synthesized at pH 9.

As mentioned by Chisti [53], using biomass after lipid extraction to produce methane by anaerobic digestion can improve the economics of microalgal biodiesel production in a refinery approach. As a substrate for anaerobic digestion, microalgal biomass with lower protein content and higher C/N ratio can diminish the ammonia toxicity and enhance methane production [54]. In this work, the highest C/N ratio of biomass produced was 15.1. The C/N ratio of lipid extracted biomass at different pH should be lower than the value shown in Table 5, because the extracted lipid is much more carbon-rich according to the lipid profile.

Table 5
TC, TN, and protein content in biomass produced at different pH in PBRs. Data are shown as means ± S.D., $n = 3$.

	pH 6.5	pH 7	pH 7.5	pH 8	pH 8.5
TC content (wt%)	48.79 ± 0.36	48.64 ± 0.28	48.18 ± 0.31	48.12 ± 0.26	48.53 ± 0.17
TN content (wt%)	3.24 ± 0.03	4.19 ± 0.07	4.62 ± 0.04	4.89 ± 0.02	6.42 ± 0.02
Protein content (wt%)	15.5 ± 0.1	20.0 ± 0.3	22.1 ± 0.2	23.4 ± 0.1	30.7 ± 0.1
C/N ratio	15.1 ± 0.2	11.6 ± 0.2	10.4 ± 0.1	9.84 ± 0.05	7.56 ± 0.03

Therefore, biomass after lipid extraction, regardless of culture pH, has a low C/N ratio compared to the optimal C/N ratio (20–35) for anaerobic digestion. A microalgal biomass co-digestion with a substrate rich in carbon is necessary for increasing the anaerobic production of methane from lipid extracted *Chlorella* biomass.

4. Conclusion

The main purpose of this work was to investigate the pH effects on cell growth, lipid production and CO_2 addition when culturing *Chlorella sorokiniana* DOE1412. By manipulating pH through CO_2 injection, pH 6 seems most beneficial for growth and lipid accumulation. At a larger scale, even though the growth rate was lower, pH 8 was more cost effective in CO_2 demand requiring 2.01 g CO_2 per g biomass. The amount of lipid was similar regardless of pH. Biodiesel produced at low pH would meet the diesel standards. The biomass made at higher pH contains more protein with a lower C/N ratio. In conclusion, this project shows how much CO_2 is required to cultivate algae in 90 L flat panel PBRs and provides further evidence that there is a balance between biomass growth, lipid production and CO_2 addition.

Acknowledgments

This work was supported through the Regional Algal Feedstock Testbed project, U.S. Department of Energy DE-EE0006269.Declaration of authors' contributions

All authors have contributed substantially to the work, and approved the final version of this manuscript. Renhe Qiu designed and performed the experiments, and wrote the manuscript. Song Gao assisted with data analysis. Paola A. Lopez helped carry out the experiments. Kimberly L. Ogden designed the experiments, supervised the project, and edited the manuscript.Conflict of interest statement

The authors declare no conflicts of interest.Statement of informed consent, human/animal rights

No conflicts, informed consent, human or animal rights applicable.

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